De La Salle University
Department of Software Technology

CCDSTRU Project Specifications
Term 2, AY 2025–2026
Due: **Mar 30, 2026 (M) before 0800**

Implement a computer program (either in C or Java) following the specifications of the system given below.

Applicable Sets

$\mathbf{C} : \{x \in \mathbf{Z}^+ \mid x < 4\}$: $\{1-3\}$

$\mathbf{N} : \{x \in \mathbf{Z}^+ \cup \{0\} \mid x <= 16\}$: $\{0-16\}$

$\mathbf{M} : \mathbf{C} \times \mathbf{C}$ 3x3 matrix $\begin{bmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix}$

$\mathbf{V} : \{\text{true}, \text{false}\}$

System Variables

$good \in \mathbf{V}$ = $\{T, F\}$

$go \in \mathbf{V}$: $\{T, F\}$

$start \in \mathbf{V}$: $\{T, F\}$

$over \in \mathbf{V}$ = $\{T, F\}$

$found \in \mathbf{V}$: $\{T, F\}$

$val \in \mathbf{N}$: $\{0-16\}$

subset of M
$R, B, S, T, F \subseteq \mathbf{M}$

$\begin{bmatrix} R & B & S \\ T & F & \end{bmatrix}$

System Facts

$F = \mathbf{M} - (R \cup B)$

$over \leftrightarrow (|F| = 3 \vee val \geq 20 \vee \neg start \wedge (|R| > 0 \wedge |B| = 0 \vee |R| = 0 \wedge |B| > 0))$

System Initialization

$good = \text{false}$

$go = \text{true}$

$start = \text{true}$

$found = \text{false}$

$val = 0$

$R = \varnothing$ null

$B = \varnothing$ null

$S = \varnothing$ null

$T = \varnothing$ null

System States and Behavior

`Remove`$(pos \in \mathbf{M})$ = $\begin{bmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix}$ pos = position in matrix?

$go \rightarrow (R = R - \{pos\})$
T

$\neg go \rightarrow (B = B - \{pos\})$
F

$S = S - \{pos\}$

$T = T - \{pos\}$

`Replace`$(pos \in \mathbf{M})$

$found = \text{false}$

$(go \wedge pos \in B) \rightarrow (B = B - \{pos\} \wedge found = \text{true})$

$(go \wedge pos \in R) \rightarrow found = \text{true}$

$(go \wedge pos \notin R) \rightarrow (R = R \cup \{pos\})$

$(\neg go \wedge pos \in R) \rightarrow (R = R - \{pos\} \wedge found = \text{true})$

$(\neg go \wedge pos \in B) \rightarrow found = \text{true}$

$(\neg go \wedge pos \notin B) \rightarrow (B = B \cup \{pos\})$

$(found \wedge pos \notin S) \rightarrow (S = S \cup \{pos\} \wedge found = \text{false})$

$(found \wedge pos \in S \wedge pos \notin T) \rightarrow (T = T \cup \{pos\} \wedge \texttt{Expand}(pos))$

$\texttt{Expand}(pos \in \mathbf{M})$

$(a, b) = pos$

$u, d, k, r \in \mathbf{M}$

$u = (a - 1, b)$

$d = (a + 1, b)$

$k = (a, b - 1)$

$r = (a, b + 1)$

$\texttt{Remove}(pos)$

$(go) \rightarrow \texttt{Replace}(u)$

$(\neg go) \rightarrow \texttt{Replace}(d)$

$\texttt{Replace}(k)$

$\texttt{Replace}(r)$

$\texttt{Update}(pos \in \mathbf{M})$

$good = \text{false}$

$(pos \notin S) \rightarrow (S = S \cup \{pos\} \wedge good = \neg good)$

$(\neg good \wedge pos \in S \wedge pos \notin T) \rightarrow (T = T \cup \{pos\} \wedge \texttt{Expand}(pos))$

$\texttt{NextPlayerMove}(pos \in \mathbf{M})$

$(\neg over \wedge start \wedge go) \rightarrow (R = R \cup \{pos\} \wedge S = S \cup \{pos\} \wedge good = \text{true})$

$(\neg over \wedge start \wedge \neg go) \rightarrow (B = B \cup \{pos\} \wedge S = S \cup \{pos\} \wedge good = \text{true})$

$(\neg over \wedge \neg start \wedge (go \wedge pos \in R \vee \neg go \wedge pos \in B)) \rightarrow (\texttt{Update}(pos) \wedge good = \text{true})$

$(start \wedge |R| = 1 \wedge |B| = 1) \rightarrow start = \text{false}$

$(\neg over \wedge good) \rightarrow (good = \neg good \wedge go = \neg go \wedge val = val + 1)$

$\texttt{GameOver}()$

$result \in \{\text{“R wins”}, \text{“B wins”}, \text{“draw”}\}$

$(over \wedge |R| > |B|) \rightarrow result = \text{“R wins”}$

$(over \wedge |R| < |B|) \rightarrow result = \text{“B wins”}$

$(over \wedge |R| = |B|) \rightarrow result = \text{“draw”}$